

Genome-wide analysis of the WRKY gene family in cotton

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Received: 26 February 2014 / Accepted: 26 May 2014 / Published online: 19 June 2014
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Abstract WRKY proteins are major transcription factors involved in regulating plant growth and development. Although many studies have focused on the functional identification of WRKY genes, our knowledge concerning many areas of WRKY gene biology is limited. For example, in cotton, the phylogenetic characteristics, global expression patterns, molecular mechanisms regulating expression, and target genes/pathways of WRKY genes are poorly characterized. Therefore, in this study, we present a genome-wide analysis of the WRKY gene family in cotton (*Gossypium*

raimondii and *Gossypium hirsutum*). We identified 116 WRKY genes in *G. raimondii* from the completed genome sequence, and we cloned 102 WRKY genes in *G. hirsutum*. Chromosomal location analysis indicated that WRKY genes in *G. raimondii* evolved mainly from segmental duplication followed by tandem amplifications. Phylogenetic analysis of alga, bryophyte, lycophyta, monocot and eudicot WRKY domains revealed family member expansion with increasing complexity of the plant body. Microarray, expression profiling and qRT-PCR data revealed that WRKY genes in *G. hirsutum* may regulate the development of fibers, anthers, tissues (roots, stems, leaves and embryos), and are involved in the response to stresses. Expression analysis showed that most group II and III *GhWRKY* genes are highly expressed under diverse stresses. Group I members, representing the ancestral form, seem to be insensitive to abiotic stress, with low expression divergence. Our results indicate that cotton WRKY genes might have evolved by adaptive duplication, leading to sensitivity to diverse stresses. This study provides fundamental information to inform further analysis and understanding of WRKY gene functions in cotton species.

Communicated by S. Hohmann.

Electronic supplementary material The online version of this article (doi:10.1007/s00438-014-0872-y) contains supplementary material, which is available to authorized users.

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Keywords WRKY transcription factor · Cotton ·
Expression profile · Development

Introduction

Plants are non-mobile; therefore, they are vulnerable to biotic stress, such as pests, fungal, bacterial and viral challenges, and abiotic stress, such as drought, cold and salinity. To adapt to, and survive in such diverse conditions, plants have developed specific responses by reprogramming their molecular, physiological and developmental processes. Transcription regulation is the key regulatory mechanism in regulating gene

expression. The WRKY protein family is one of the largest families of transcription regulators in plants, and is named after its characteristic protein sequence. Since the first *WRKY* genes were cloned from sweet potato (Ishiguro and Nakamura 1994), more *WRKY* genes have been cloned from other plants. To date, *WRKY* proteins have been identified only in plants. In recent years, interest in *WRKY* transcription factors has increased, not only because of their numerous members and diverse functions in plant development, but also because of their value in evolutionary research.

WRKY transcription factors bind specific DNA sequences to activate or repress transcription of multiple target genes (Yang and Chen 2001). The conserved *WRKY* domain contains approximately 60 amino acid residues. In the *WRKY* domain, a conserved *WRKYGQK* hexapeptide sequence is usually followed by a C2H2- or C2HC-type zinc finger motif. *WRKY* transcription factors are classified according to the number of *WRKY* domains and the zinc finger motif that they contain: group I members have two *WRKY* domains; whereas group II and group III members have only a single *WRKY* domain, followed by a novel zinc-finger-like motif C2H2 (C-X₄₋₅-C-X₂₂₋₂₃-H-X-H) and C2HC (C-X₇-C-X₂₃-H-X-C), respectively (Eulgem et al. 2000).

Plant *WRKY* transcription factors are involved in various physiological processes, such as *Arabidopsis* seed coat and trichome development (Johnson et al. 2002), somatic embryogenesis of cock's foot, *Dactylis glomerata* (Alexandrova and Conger 2002), the gibberellin signaling pathway in aleurone cells of rice, *Oryza sativa* (Zhang et al. 2004) and leaf senescence in *Arabidopsis* (Miao et al. 2004). In addition, *WRKY* genes respond to many biotic and abiotic stresses, including pathogenic bacteria (Zheng et al. 2006), wounding (Hara et al. 2000), oxidative stress (Rizhsky et al. 2004), drought, salinity, heat (Pnueli et al. 2002) and freezing (Huang and Duman 2002). To date, the *WRKY* gene family has been analyzed in many plants, including *Arabidopsis thaliana* and *Arabidopsis lyrata* (Song and Gao 2014), *Cucumis sativus* (Ling et al. 2011), *Oryza sativa* (Ross et al. 2007; Ramamoorthy et al. 2008) and *Populus trichocarpa* (He et al. 2012).

Cotton (*Gossypium* spp.) is an important economic crop and a model plant for the study of polyploids, cell elongation and cell wall synthesis (Paterson et al. 2012). *G. hirsutum* (AD₁, 2n = 4x = 52), the allotetraploid species, is the most widely planted cotton species in the world, accounting for 90 % of all cotton production (Soltis et al. 2004). The allotetraploid species was derived from the D-genome species, *G. raimondii*, as the pollen-providing parent, and an A-genome species, *G. arboreum*, as the maternal parent (Sunilkumar et al. 2006). They diverged 5–10 million years ago (Senchina et al. 2003), and reunited 1–2 million years ago forming *G. hirsutum* (Paterson et al. 2012).

In cotton, *GhWRKY15* was significantly induced in seedlings following fungal infection or treatment with salicylic acid (SA), methyl jasmonate or methyl viologen (Yu et al. 2012). *GhWRKY3* was constitutively expressed in roots, stems and leaves; it was upregulated by the application of various phytohormones, including SA, methyl jasmonate (MeJA), abscisic acid (ABA), gibberellins (GAs) and ethylene (ET), and showed enhanced expression after infection with *Rhizoctonia solani*, *Colletotrichum gossypii* and *Fusarium oxysporum* f. sp. *vasinfectum* (Guo et al. 2011). *GhWRKY1* was a pathogen-inducible transcription factor and played an important role in plant defense responses (Xu et al. 2012). *GhWRKY40* was a stress-inducible transcription factor, and played an important role in response to wounding and *Ralstonia solanacearum* infection (Wang et al. 2014). Moreover, Cai et al. reported basic information of the genome-wide *WRKY* gene family in *G. raimondii* (Cai et al. 2014).

The *G. raimondii* genome has been sequenced and this provides the opportunity to perform a genome-wide analysis of *WRKY* genes in cotton (*G. raimondii* and *G. hirsutum*). We identified 116 *GrWRKY* genes in the *G. raimondii* genome, and cloned 102 *GhWRKY* sequences from *G. hirsutum* cDNA by homology-based cloning methods based on the *G. raimondii* genes. The *GhWRKY* genes analyzed represent only part of the *WRKY* family in *G. hirsutum*. To better understand the relationship between *GrWRKYs* and *GhWRKYs*, we examined the distribution of *GrWRKYs* on chromosomes, and inferred that the *GrWRKYs* mainly evolved from segmental duplication followed by tandem amplifications. Simple sequence repeats (SSRs) could separate the *GrWRKYs* and *GhWRKYs* into three groups. To determine the evolutionary relationship among different subgroups, we downloaded *WRKY* protein sequences from algae, bryophytes, lycophyta, monocots and eudicots and performed a phylogenetic analysis. To understand the expression patterns of *GhWRKYs* in different development stages and tissues and in response to abiotic stress, we evaluated their expressions using publicly available microarray and expression profiling data, and performed quantitative reverse transcriptase PCR (qRT-PCR). Our results provide valuable information about *WRKY* genes in *G. raimondii* and *G. hirsutum* that will help future studies of evolutionary relationships among cotton species.

Materials and methods

Identification and annotation of the *WRKY* gene family in cotton

We downloaded *G. raimondii* genome sequences (G.raimondii.chromosome.fasta.gz) and the proteome

sequence (G.raimondii.pep.fasta.gz) from the Cotton Genome Project (CGP; <http://cgp.genomics.org.cn/page/species/index.jsp>). A local BLASTP search was performed to identify complete WRKY members, using *Arabidopsis* WRKY protein sequences as query sequences. The *e* value for BLASTP was set at $1e-10$ to obtain the final dataset of WRKY proteins. Overlapping and non-targeted protein sequences were manually removed, resulting in 116 WRKY protein sequences. To identify WRKY genes in *G. hirsutum*, primers were designed according to *GrWRKY* sequences, which were used with mixed cDNA from roots, stems, leaves, flowers and anthers of *G. hirsutum* cultivar CCRI10 as a template for PCR. The PCR products were ligated into the T-vector, and sequenced by GENEWIZ Sequencing Services (NJ, USA).

Online software hosted by the Cotton Marker Database (http://www.cottonmarker.org/cgi-bin/cmd_ssr) was used to identify simple sequence repeats (SSRs) in cotton WRKY gene family members. SSR motifs, with repeat units of more than six for dinucleotides, four for trinucleotides, and three for tetranucleotides, pentanucleotides, and hexanucleotides were used as the search criteria (Lai et al. 2011).

Mapping WRKY genes on cotton chromosomes

A local blast search of the *G. raimondii* genome sequence was performed to map the physical location of the 116 genes. The Mapchart 2.2 software was used to visualize the distribution of the WRKY genes on the 13 *G. raimondii* chromosomes.

Phylogenetic analysis of the WRKY gene family

WRKY family sequences from rice (*Oryza sativa*; japonica cultivar-group), maize (*Zea mays*) (Reik and Walter 2001) and *Cyanidioschyzon merolae* (Matsuzaki et al. 2004) were downloaded from the NCBI. *Arabidopsis* sequences were retrieved from TAIR (<http://www.arabidopsis.org/>). Sequences from *Scherffelia dubia*, *Chlorella* sp. NC64A, *Chlamydomonas reinhardtii*, *Volvox carteri*, *Physcomitrella patens*, *Selaginella moellendorffii* and *Populus trichocarpa* were downloaded from the PlnTFDB (<http://plntfdb.bio.uni-potsdam.de/v3.0/>).

Phylogenetic and molecular evolutionary analysis was conducted using MEGA version 5.2 (<http://www.megasoftware.net/>) with pairwise distance and the neighbor-joining algorithm. The p-distance method was used to compute the evolutionary distances, which were used to estimate the number of amino acid substitutions per site. Conducting 1,000 bootstrap sampling steps (Wei et al. 2012) established the reliability of each tree.

Expression analysis methods

Public cotton expression datasets were obtained from the Plant Expression Database (PLEXdb) and the Gene Expression Omnibus (GEO). In addition, cotton microarray-based datasets from cotton fiber development stages at 0, 6, 9, 12, 19 and 25 days post anthesis (dpa) from JKC725, JKC703, JKC777, JKC783 and JKC737 varieties of *G. hirsutum* (GEO accession number GSE36228) were obtained (Nigam and Sawant 2013). The accession number of the waterlog-stressed *G. hirsutum* root and leaf tissues data is GSE16467 (Christianson et al. 2010). The accession number of the microarray data from field-grown drought-stressed *G. hirsutum* leaf is GSE18253 (Cottee et al. 2014). The accession number of *G. hirsutum* leaf tissue under drought stress is GSE29566 (Padmalatha et al. 2012). The accession number of microarray analysis of *G. hirsutum* under drought stress during fiber development stages at 0, 5, 10, 15 and 20 dpa is GSE29567 (Padmalatha et al. 2012). The accession number of the microarray analysis in leaf tissue and during fiber development stages at 0, 5, 10, 15 and 20 dpa of *G. hirsutum* under drought stress is GSE29810 (Nigam and Sawant 2013). The accession number of *G. hirsutum* under ABA, drought, alkalinity, cold, and salinity stress is GSE50770 (Zhu et al. 2013). The robust multichip analysis (RMA) algorithm was used to normalize all microarray data used.

The RNA-Seq data of the leaf senescence process in *G. hirsutum* in 15-, 25-, 35-, 45-, 55-, and 65-day-old leaves were used. The accession number of the reference genes is SSR654704, and the accession number of the expression profiling data is SRA656612. The transcript per million clean tags (TPM) algorithm was used to normalize the data. RNA-Seq data of gene expression during anther development at tetrad pollen (TTP), uninucleate pollen (UNP), binucleate pollen (BNP) and mature pollen (MTP) stages, and in roots, stems, leaves and embryos in *G. hirsutum* were used. The reads per kilobase of exon model per million mapped reads (RPKM) algorithm was used to normalize these data. Expression profiling data and their reference genes can be downloaded from the supplementary data of published papers (Ma et al. 2012, 2013); the doi numbers of the two papers are doi: 10.1371/journal.pone.0049244 and doi:10.1111/jipb.12067.

qRT-PCR

To examine gene expression during fiber development, fiber samples of TM-1 at 0, 5, 10, 15, 20 and 25 dpa stages were subjected to qRT-PCR; each sample was obtained from approximately 25 individual plants for each stage. An RNApure pure plant kit (TIANGEN, China) was used to extract the total RNA. Reverse transcription reactions were

performed using 2.0 µg RNA with SuperScript III reverse transcriptase (Invitrogen, USA). Primers were designed using Oligo7 and synthesized by GENEWIZ. Reactions were carried out using SYBR Green PCR Master Mix (Roche Applied Science, Germany) on an ABI7500 real-time PCR system (Applied Biosystems, USA) with triplicate technical replicates. The amplification of *GhHis3* was used as the reference to normalize the qRT-PCR data (Tu et al. 2007). Reaction volumes of 20 µl contained 10 µl SYBR Green PCR Master Mix, 7.2 µl distilled H₂O, 0.8 µl primers and 2 µl cDNA. Amplification reactions were initiated with a pre-denaturing step (95 °C, 10 min), followed by denaturing (95 °C, 10 s) and annealing (60 °C, 30 s) for 40 cycles. Data were processed using the $2^{-\Delta\Delta C_t}$ method (Livak and Schmittgen 2001).

Results

Identification of the WRKY gene family in cotton

In the *G. raimondii* genome, 116 genes were identified as possible members of the WRKY gene family. Of these, 113 genes had intact WRKY domain structures. Eighteen proteins had been annotated previously and were revealed to have two complete WRKY domains each (Table 1). One hundred and seven WRKY genes could be mapped onto chromosomes and were named based on their order on the chromosomes (from chromosome 1 to 13) as *GrWRKY1* to *GrWRKY107* (Fig. 1). Nine WRKY genes (Cotton_D_gene_10026362, Cotton_D_gene_10006792, Cotton_D_gene_10030213, Cotton_D_gene_10015040, Cotton_D_gene_10001713, Cotton_D_gene_10000655, Cotton_D_gene_10015108, Cotton_D_gene_10015628, and Cotton_D_gene_10016391) could not be conclusively mapped to any chromosome, and were named *GrWRKY108-GrWRKY116*, respectively (Fig. 1).

WRKY genes were scattered throughout the *G. raimondii* chromosomes. Chromosome 7 had the largest number of *GrWRKY* genes (17, 16.04 %), whereas chromosome 5 had the fewest (3, 2.83 %). Chromosomes 9 and 11 contained only group II WRKY genes. Chromosomes 2, 3 and 6 contained only group I and II members. Chromosome 5 had no group I WRKY genes. This unbalanced distribution of *GrWRKY* genes on chromosomes suggested that genetic variations existed in the evolutionary process.

Comparing the *GrWRKY* protein domain structures with those of *AtWRKY* proteins confirmed variations in the WRKY domain structure. These variations occurred in the following WRKY domains: WRKY-C-X4-C (*GrWRKY94*), WRKY-CX (*GrWRKY66*), WRKY-X (*GrWRKY9*, 20, 34, 55 and 45) and WRKGYR (*GrWRKY72*).

To establish whether *GrWRKY* genes existed or were expressed in *G. hirsutum*, we first designed primers from the 3'-untranslated region (UTR) and 5'-UTR of the *GrWRKY* genes. These were used to clone homologous sequences from *G. hirsutum* cDNAs. We identified 102 of the annotated *GrWRKY* genes in *G. hirsutum* (Table 1), which were named as *GhWRKY* genes, each gene using the same suffix number as its *GrWRKY* ortholog. Two of these *GrWRKY* genes already existed in the NCBI database; the remaining 100 cloned cDNAs were submitted to NCBI, and their accession numbers are provided in Table 1. The primer sequences used for cDNA cloning are listed in Supplementary data S3.

Evolutionary analysis of the WRKY transcription factor family

To study the evolutionary origin of the WRKY gene family in cotton, we used intact WRKY domain sequences derived from the complete genomes of red algae (*C. merolae*), glaucophyta (*G. nostochinearum* and *C. paradoxa*), chlorophyta (*S. dubia*, *Chlorella* sp. NC64A, *C. reinhardtii*, and *V. carteri*), bryophytes (*P. patens*), lycophyta (*S. moellendorffii*), monocots (*Z. mays*, and *Oryza sativa*) and eudicots (*P. trichocarpa*, *A. thaliana*, *G. raimondii*, and *G. hirsutum*). Considering the high similarity between the *GrWRKY* and *GhWRKY* proteins, we used the *GhWRKY* protein domains to represent cotton in the phylogenetic tree.

There are 30 or fewer WRKY proteins in each species of the early diverged plants, algae, bryophytes and lycophyta. In contrast, over 100 WRKY proteins have been identified in *Z. mays* (136), *O. sativa* (102), *P. trichocarpa* (104), and we identified 116 in *G. raimondii* (Supplementary Fig. 1). This indicated that the number of WRKY gene family members increased during the evolution from algae to angiosperms. This expansion co-evolved with the increasing complexity of the plant body, and suggests that environmental pressures drove the WRKY gene family expansion to adapt to environmental changes.

The difference between the N- and C-terminals of group I WRKY domains prompted us to classify them into two independent domains, IN and IC. A phylogenetic tree was created according to the alignment results (Fig. 2). This revealed a well-organized classification in accordance with *Arabidopsis* (Eulgem et al. 2000) and maize (Nagata et al. 2012), and the clades were named as group IN, IC, IIa, IIb, IIc, IId, IIe and III.

In Fig. 2, group II members form three distinct clades: group IIa + IIb, IIc and group IId + IIe. Moreover, group IIa + IIb and group IIc are closely related to group IC. Meanwhile, group IId + group IIe are closely related to group III. Group IIa and IIb form a large, non-monophyletic

Table 1 The WRKY transcription factor family in cotton

Gossypium raimondii					Gossypium hirsutum					
Gene name	Gene ID	ORF length	SSR site	Subgroup	Gene name	Accession no.	ORF length	SSR site	Subgroup	Identity (%)
GrWRKY1	Cotton_D_gene_10023185	930	2	IIa	GhWRKY1	KF669831	930	1	IIa	98.5
GrWRKY2	Cotton_D_gene_10019327	483	0	IIc	GhWRKY2	KF669759	483	0	IIc	99.79
GrWRKY3	Cotton_D_gene_10009963	1,548	0	I	GhWRKY3	KF669771	1,548	1	I	98.58
GrWRKY4	Cotton_D_gene_10026164	1,680	2	IIb	GhWRKY4	KF669822	1,683	0	IIb	82.35
GrWRKY5	Cotton_D_gene_10028479	993	0	III	GhWRKY5	KF669781	1,005	0	III	99
GrWRKY6	Cotton_D_gene_10028569	1,086	0	IIb	GhWRKY6	KF669821	1,086	0	IIb	99.54
GrWRKY7	Cotton_D_gene_10016255	915	0	III	GhWRKY7	KF669776	915	0	III	99.46
GrWRKY8	Cotton_D_gene_10023049	1,545	1	IIb	GhWRKY8	KF669823	1,287	0	IIb	81.57
GrWRKY9	Cotton_D_gene_10022823	330	1	IIc	GhWRKY9	KF669841	483	1	IIc	98.7
GrWRKY10	Cotton_D_gene_10015280	1,476	3	I	GhWRKY10	KF669760	1,515	3	I	98.05
GrWRKY11	Cotton_D_gene_10015331	912	0	IIa	GhWRKY11	KF669832	912	0	IIa	97.38
GrWRKY12	Cotton_D_gene_10012784	1,005	0	IIId	GhWRKY12	KF669853	1,005	0	IIId	97.33
GrWRKY13	Cotton_D_gene_10033299	1,323	1	IIb	GhWRKY13	—	—	—	—	—
GrWRKY14	Cotton_D_gene_10006909	1,683	1	I	GhWRKY14	KF669762	1,710	2	I	99.13
GrWRKY15	Cotton_D_gene_10020058	450	0	IIc	GhWRKY15	KF669833	450	0	IIc	98.23
GrWRKY16	Cotton_D_gene_10024898	1,575	1	IIb	GhWRKY16	KF669824	1,575	1	IIb	98.86
GrWRKY17	Cotton_D_gene_10024943	1,704	1	I	GhWRKY17	KF669761	1,701	1	I	98.17
GrWRKY18	Cotton_D_gene_10012177	1,719	0	I	GhWRKY18	KF669858	1,719	0	I	100
GrWRKY19	Cotton_D_gene_10031091	837	0	IIe	GhWRKY19	KF669784	792	0	IIe	97.36
GrWRKY20	Cotton_D_gene_10036575	381	0	IIc	GhWRKY20	KF669806	345	1	IIc	100
GrWRKY21	Cotton_D_gene_10036580	645	0	IIc	GhWRKY21	KF669807	645	0	IIc	99.07
GrWRKY22	Cotton_D_gene_10002745	1,668	1	I	GhWRKY22	KF669763	1,206	1	I	98.92
GrWRKY23	Cotton_D_gene_10007968	1,020	1	IIId	GhWRKY23	KF669794	1,014	1	IIId	98.04
GrWRKY24	Cotton_D_gene_10002760	2,175	0	I	GhWRKY24	KF669764	2,175	0	I	99.49
GrWRKY25	Cotton_D_gene_10029195	2,037	0	IIc	GhWRKY25	KF669808	2,037	0	IIc	97.89
GrWRKY26	Cotton_D_gene_10027942	1,662	0	IIb	GhWRKY26	—	—	—	—	—
GrWRKY27	Cotton_D_gene_10007498	1,065	0	III	GhWRKY27	KF669775	1,065	0	III	99.63
GrWRKY28	Cotton_D_gene_10009592	876	0	IIId	GhWRKY28	KF669796	876	0	IIId	97.04
GrWRKY29	Cotton_D_gene_10005114	873	0	IIId	GhWRKY29	KF669795	351	0	IIId	100
GrWRKY30	Cotton_D_gene_10005113	969	0	IIId	GhWRKY30	KF669856	903	0	IIId	100

Table 1 continued

Gossypium raimondii					Gossypium hirsutum					
Gene name	Gene ID	ORF length	SSR site	Subgroup	Gene name	Accession no.	ORF length	SSR site	Subgroup	Identity (%)
GrWRKY31	Cotton_D_gene_10002578	921	2	III	GhWRKY31	KF669773	918	2	III	97.19
GrWRKY32	Cotton_D_gene_10005791	1,044	2	IIc	GhWRKY32	KF669809	1,044	2	IIc	99.04
GrWRKY33	Cotton_D_gene_10037859	897	1	IIc	GhWRKY33	KF669810	891	1	IIc	97.33
GrWRKY34	Cotton_D_gene_10024223	372	0	IC	GhWRKY34	KF669842	360	0	IC	100
GrWRKY35	Cotton_D_gene_10014412	1,221	0	IIb	GhWRKY35	—	—	—	—	—
GrWRKY36	Cotton_D_gene_10027087	1,539	0	I	GhWRKY36	FJ966887 ^a	1,545	0	I	97.7
GrWRKY37	Cotton_D_gene_10027029	903	0	IIc	GhWRKY37	KF669811	903	0	IIc	98.79
GrWRKY38	Cotton_D_gene_10017279	2,697	0	IIc	GhWRKY38	KF669838	2,859	0	IIc	99.25
GrWRKY39	Cotton_D_gene_10021341	876	1	IIc	GhWRKY39	KF669812	780	1	IIc	99.27
GrWRKY40	Cotton_D_gene_10012245	1,455	0	I	GhWRKY40	KF669767	1,455	0	I	99.18
GrWRKY41	Cotton_D_gene_10019857	921	0	IIc	GhWRKY41	—	—	—	—	—
GrWRKY42	Cotton_D_gene_10036922	1,029	2	IIId	GhWRKY42	KF669797	1,035	3	IIId	97.21
GrWRKY43	Cotton_D_gene_10004689	1,698	2	IIb	GhWRKY43	—	—	—	—	—
GrWRKY44	Cotton_D_gene_10007835	588	0	IIc	GhWRKY44	—	—	—	—	—
GrWRKY45	Cotton_D_gene_10020122	504	1	IIc	GhWRKY45	KF669840	504	2	IIc	97.65
GrWRKY46	Cotton_D_gene_10014637	1,209	0	I	GhWRKY46	KF669766	1,212	0	I	99.09
GrWRKY47	Cotton_D_gene_10009679	1,008	0	IIId	GhWRKY47	KF669798	1,008	0	IIId	98.51
GrWRKY48	Cotton_D_gene_10035636	1,005	0	IIe	GhWRKY48	KF669785	1,005	0	IIe	99.31
GrWRKY49	Cotton_D_gene_10035639	567	1	IIc	GhWRKY49	KF669813	567	1	IIc	99.12
GrWRKY50	Cotton_D_gene_10035678	996	0	III	GhWRKY50	KF669783	1,002	0	III	98.41
GrWRKY51	Cotton_D_gene_10035779	1,515	1	IIb	GhWRKY51	KF669825	1,518	1	IIb	97.37
GrWRKY52	Cotton_D_gene_10035819	936	0	IIc	GhWRKY52	KF669850	672	1	IIc	98.36
GrWRKY53	Cotton_D_gene_10035830	801	0	IIe	GhWRKY53	KF669786	801	0	IIe	98.76
GrWRKY54	Cotton_D_gene_10008065	966	2	IIId	GhWRKY54	KF669799	972	2	IIId	98.56
GrWRKY55	Cotton_D_gene_10008614	780	1	IIId	GhWRKY55	GU207869 ^a	780	0	IIId	99.11
GrWRKY56	Cotton_D_gene_10017802	996	2	III	GhWRKY56	KF669779	996	2	III	99.3
GrWRKY57	Cotton_D_gene_10017815	1,047	1	IIe	GhWRKY57	KF669787	1,047	1	IIe	97.14
GrWRKY58	Cotton_D_gene_10030785	1,188	4	IIe	GhWRKY58	KF669788	1,188	3	IIe	97.9
GrWRKY59	Cotton_D_gene_10030817	885	0	III	GhWRKY59	KF669782	858	0	III	99.17
GrWRKY60	Cotton_D_gene_10016858	1,065	0	III	GhWRKY60	KF669778	1,065	0	III	99.34

Table 1 continued

Gossypium raimondii					Gossypium hirsutum					
Gene name	Gene ID	ORF length	SSR site	Subgroup	Gene name	Accession no.	ORF length	SSR site	Subgroup	Identity (%)
GrWRKY61	Cotton_D_gene_10016888	897	0	Ile	GhWRKY61	KF669790	900	0	Ile	97.34
GrWRKY62	Cotton_D_gene_10025638	825	0	Ile	GhWRKY62	—	—	—	—	—
GrWRKY63	Cotton_D_gene_10029943	4,587	0	I	GhWRKY63	—	—	—	—	—
GrWRKY64	Cotton_D_gene_10029945	2,226	0	IC	GhWRKY64	KF669837	609	0	IC	99.47
GrWRKY65	Cotton_D_gene_10027229	627	1	Ile	GhWRKY65	KF669789	686	0	Ile	97
GrWRKY66	Cotton_D_gene_10014946	537	2	Ile	GhWRKY66	KF669848	537	2	Ile	99.26
GrWRKY67	Cotton_D_gene_10014899	927	0	Ile	GhWRKY67	KF669844	927	0	Ile	99.46
GrWRKY68	Cotton_D_gene_10006708	1,821	5	Ile	GhWRKY68	KF669826	1,824	4	Ile	98.47
GrWRKY69	Cotton_D_gene_10037032	990	0	Ile	GhWRKY69	KF669845	990	0	Ile	98.39
GrWRKY70	Cotton_D_gene_10037078	756	0	Ile	GhWRKY70	KF669834	756	1	Ile	98.15
GrWRKY71	Cotton_D_gene_10037079	939	1	Ile	GhWRKY71	KF669857	939	1	Ile	98.62
GrWRKY72	Cotton_D_gene_10003094	1,338	1	Ile	GhWRKY72	KF669791	1,377	1	Ile	99.73
GrWRKY73	Cotton_D_gene_10007125	930	1	Ile	GhWRKY73	KF669835	924	1	Ile	97.43
GrWRKY74	Cotton_D_gene_10033628	957	0	Ile	GhWRKY74	KF669814	957	0	Ile	97.81
GrWRKY75	Cotton_D_gene_10033661	831	0	Ile	GhWRKY75	—	—	—	—	—
GrWRKY76	Cotton_D_gene_10033857	1,695	0	Ile	GhWRKY76	KF669827	1,692	0	Ile	98.4
GrWRKY77	Cotton_D_gene_10023486	513	0	IC	GhWRKY77	KF669816	513	0	IC	97.87
GrWRKY78	Cotton_D_gene_10023655	1,062	0	Ile	GhWRKY78	KF669801	1,062	0	Ile	98.69
GrWRKY79	Cotton_D_gene_10035431	1,482	1	Ile	GhWRKY79	KF669828	1,524	1	Ile	99.37
GrWRKY80	Cotton_D_gene_10035475	945	2	Ile	GhWRKY80	KF669815	951	2	Ile	98.53
GrWRKY81	Cotton_D_gene_10025883	1,326	1	Ile	GhWRKY81	KF669830	1,326	1	Ile	98.64
GrWRKY82	Cotton_D_gene_10009488	1,488	2	Ile	GhWRKY82	KF669768	1,113	2	Ile	98.42
GrWRKY83	Cotton_D_gene_10039495	936	1	I	GhWRKY83	KF669836	936	1	I	98.23
GrWRKY84	Cotton_D_gene_10039553	1,050	1	Ile	GhWRKY84	KF669802	1,023	1	Ile	99.25
GrWRKY85	Cotton_D_gene_10039598	828	0	Ile	GhWRKY85	KF669792	828	0	Ile	99.32
GrWRKY86	Cotton_D_gene_10040769	450	1	Ile	GhWRKY86	KF669817	450	1	Ile	99.28
GrWRKY87	Cotton_D_gene_10040730	1,653	0	Ile	GhWRKY87	KF669829	1,683	1	Ile	98.01
GrWRKY88	Cotton_D_gene_10035091	828	0	Ile	GhWRKY88	KF669774	825	2	Ile	99.46
GrWRKY89	Cotton_D_gene_10005482	1,131	0	III	GhWRKY89	—	—	—	—	—
GrWRKY90	Cotton_D_gene_10031158	1,638	2	Ile	GhWRKY90	KF669851	1,509	0	Ile	97

Table 1 continued

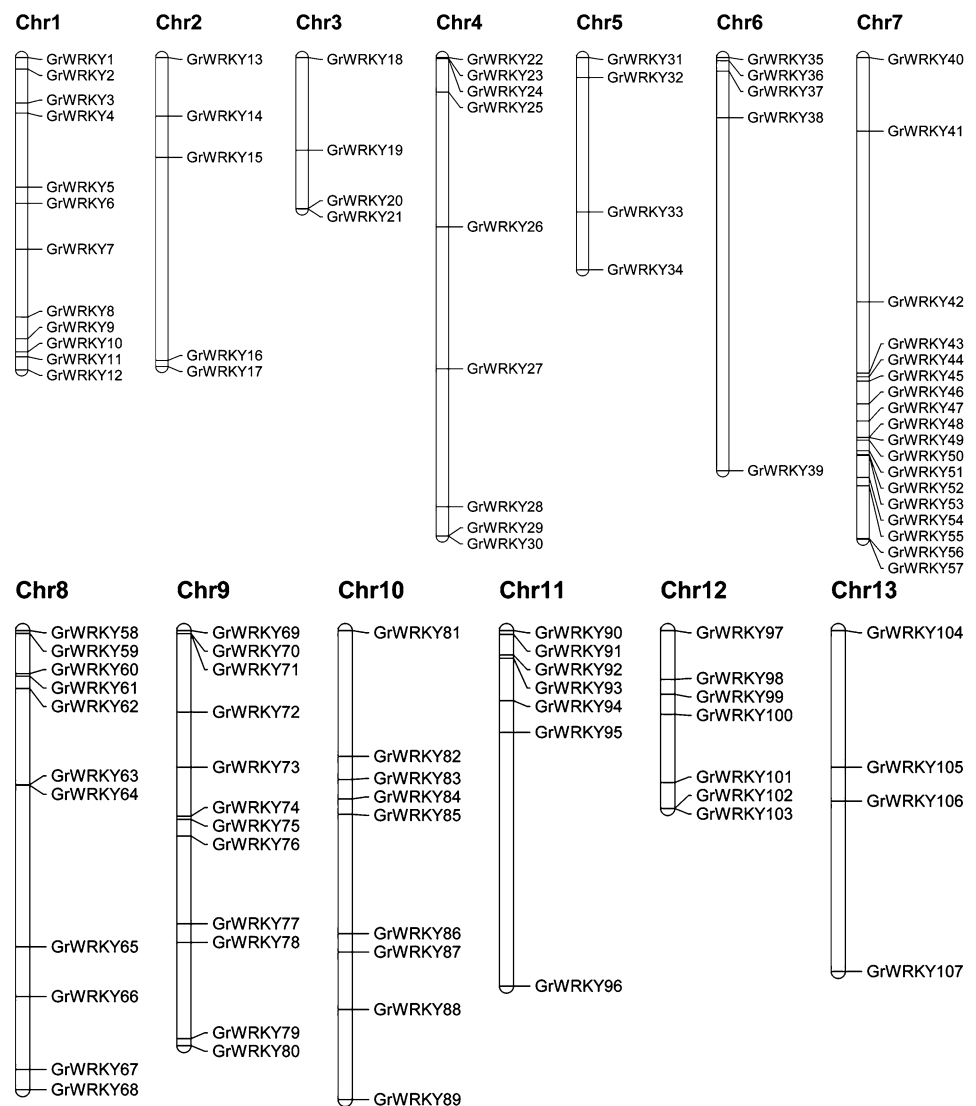
Gossypium raimondii					Gossypium hirsutum					
Gene name	Gene ID	ORF length	SSR site	Subgroup	Gene name	Accession no.	ORF length	SSR site	Subgroup	Identity (%)
GrWRKY91	Cotton_D_gene_10031219	822	3	IIb	GhWRKY91	KF669793	819	4	IIb	98.85
GrWRKY92	Cotton_D_gene_10031462	942	1	IIe	GhWRKY92	KF669849	942	1	IIe	98.79
GrWRKY93	Cotton_D_gene_10031500	1,074	1	IIc	GhWRKY93	KF669854	1,074	1	IIc	98.73
GrWRKY94	Cotton_D_gene_10011482	363	0	IId	GhWRKY94	KF669847	480	0	IId	98.76
GrWRKY95	Cotton_D_gene_10010798	1,038	0	IC	GhWRKY95	KF669855	1,080	0	IC	97.44
GrWRKY96	Cotton_D_gene_10015718	1,500	2	IIc	GhWRKY96	KF669769	1,509	2	IIc	98.39
GrWRKY97	Cotton_D_gene_10016568	1,398	2	I	GhWRKY97	KF669852	1,401	2	I	98.78
GrWRKY98	Cotton_D_gene_10027770	891	0	I	GhWRKY98	KF669818	714	1	I	99.21
GrWRKY99	Cotton_D_gene_10026013	558	0	IIc	GhWRKY99	KF669846	558	0	IIc	99.44
GrWRKY100	Cotton_D_gene_10001752	954	0	IIc	GhWRKY100	KF669819	963	0	IIc	97.28
GrWRKY101	Cotton_D_gene_10019812	1,077	1	IIc	GhWRKY101	KF669777	1,080	1	IIc	97.2
GrWRKY102	Cotton_D_gene_10016710	906	2	III	GhWRKY102	KF669772	906	2	III	99.08
GrWRKY103	Cotton_D_gene_10016711	837	1	III	GhWRKY103	KF669820	837	1	III	98.24
GrWRKY104	Cotton_D_gene_10022405	1,800	0	IIc	GhWRKY104	—	—	—	—	—
GrWRKY105	Cotton_D_gene_10033488	1,455	3	IIb	GhWRKY105	KF669770	1,203	0	IIb	97.3
GrWRKY106	Cotton_D_gene_10033582	1,475	0	I	GhWRKY106	KF669780	915	0	I	97.9
GrWRKY107	Cotton_D_gene_10024748	915	0	III	GhWRKY107	—	—	—	—	—
GrWRKY108	Cotton_D_gene_10026362	1,362	0	IIc	GhWRKY108	KF669765	1,362	0	IIc	98.73
GrWRKY109	Cotton_D_gene_10006792	2,193	0	I	GhWRKY109	—	—	—	—	—
GrWRKY110	Cotton_D_gene_10030213	2,289	0	IIc	GhWRKY110	—	—	—	—	—
GrWRKY111	Cotton_D_gene_10015040	993	0	IIc	GhWRKY111	KF669800	993	0	IIc	99.36
GrWRKY112	Cotton_D_gene_10001713	471	0	I	GhWRKY112	KF669803	471	0	I	97
GrWRKY113	Cotton_D_gene_10000655	621	1	I	GhWRKY113	KF669804	621	0	I	97.1
GrWRKY114	Cotton_D_gene_10015108	225	0	IIc	GhWRKY114	KF669805	486	0	IIc	98.36
GrWRKY115	Cotton_D_gene_10015628	570	0	IIc	GhWRKY115	KF669839	522	0	IIc	99.39
GrWRKY116	Cotton_D_gene_10016391	486	0	IId	GhWRKY116	KF669843	666	0	IId	99.5

WRKY transcription factors in *Gossypium raimondii* and *Gossypium hirsutum*. We named the *GrWRKY* genes according to their location on chromosomes, and named the *GhWRKY* genes according to their *GrWRKY* orthologs. Gene IDs are from the *Gossypium raimondii* genome sequence database. Accession no. is the accession number of the *GhWRKY* genes in NCBI. Subgroups were classified according to BLAST searches of the *GhWRKY* and *GrWRKY* proteins with *AtWRKY* domains. Identity is the result of BLASTn searching between *GhWRKY* and *GrWRKY* sequences

— No result

^a Already exist in NCBI

Fig. 1 Mapping of the WRKY gene family members to *Gossypium raimondii* chromosomes. The putative WRKY genes were named as *GrWRKY1* to *GrWRKY107*, based on their order on the chromosomes 1–13 and from top to bottom



subtree with two distinct clades. Interestingly, no alga, bryophyte or lycophyta WRKY domains were classified as group IIa. This phenomenon may indicate that group IIa evolved from group IIb. Similarly, we could infer that group IIe evolved from group IId. Most of the algal genes were classified into group IN and IC, while some algal proteins that have single WRKY domains were also classified into group IC. Domains from algae, lycophyta and bryophytes were mainly classified into IC, IIc, IIb, IN, IId and III, which implied that groups IIb, IIc, IId and III may have evolved from group I. Just two *S. moellendorffii* WRKY proteins were classified as group III; the appearance of this group predates the divergence of lycophyta/angiosperms.

WRKY genes from the same taxonomic class tended to cluster together in the phylogenetic tree (Kumar et al. 2011) and were not equally represented within a given clade, suggesting that they had experienced duplications after the plant classes diverged. Most genes from *G. hirsutum* and *G.*

raimondii share high similarity sequences, with identities ranging from 81.57 to 100 % (Table 1). Therefore, SSRs could be used to clarify their differences (Hou et al. 2014).

Figure 3 shows the frequency of SSR motifs in the WRKY genes of *G. hirsutum* and *G. raimondii*. These sequences could be classified into three groups. The first group consists of CCG/CGG, ATC/GAT, TTC/GAA, AAC/GTT, ACA/TGT, ATG/CAT and CTC/GAG, which were more frequent in *G. raimondii*. The second group contains CAG/CTG, ACC/GGT, AGA/TCT, TGC/GCA, CGC/GCG, TCC/GGA, AGG/CCT, AGT/ACT, ATA/TAT and TTGG/CCTT; these were found at similar frequencies in both *G. raimondii* and *G. hirsutum*. The third group includes AAG/CTT, AAT/ATT, AGC/GCT, AT/TA, CAC/GTG, AG/CT, CCA/TGG, CAA/TTG and TGA/TCA, which were the predominant motifs in *G. hirsutum*. These motif groupings will be useful for *Gossypium* genus evolutionary analysis (Hou et al. 2014).

Fig. 2 Unrooted neighbor-joining phylogenetic tree of WRKY domains in the investigated plants. The tree was constructed using intact WRKY domain sequences from the complete WRKY gene families of red algae (*Cyanidioschyzon merolae*); the glaucophyta (*Scherffelia dubia*, *Chlorella* sp. NC64A, *Chlamydomonas reinhardtii* and *Volvox carteri*); the bryophytes (*Physcomitrella patens*); lycopphyta (*Selaginella moellendorffii*); the monocots (*Zea mays* and *Oryza sativa*); and the eudicots (*Populus trichocarpa*, *Arabidopsis thaliana* and *Gossypium hirsutum*)

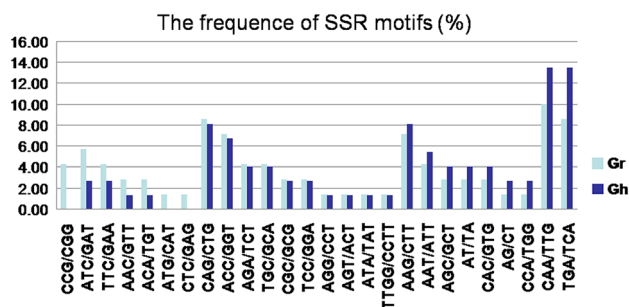
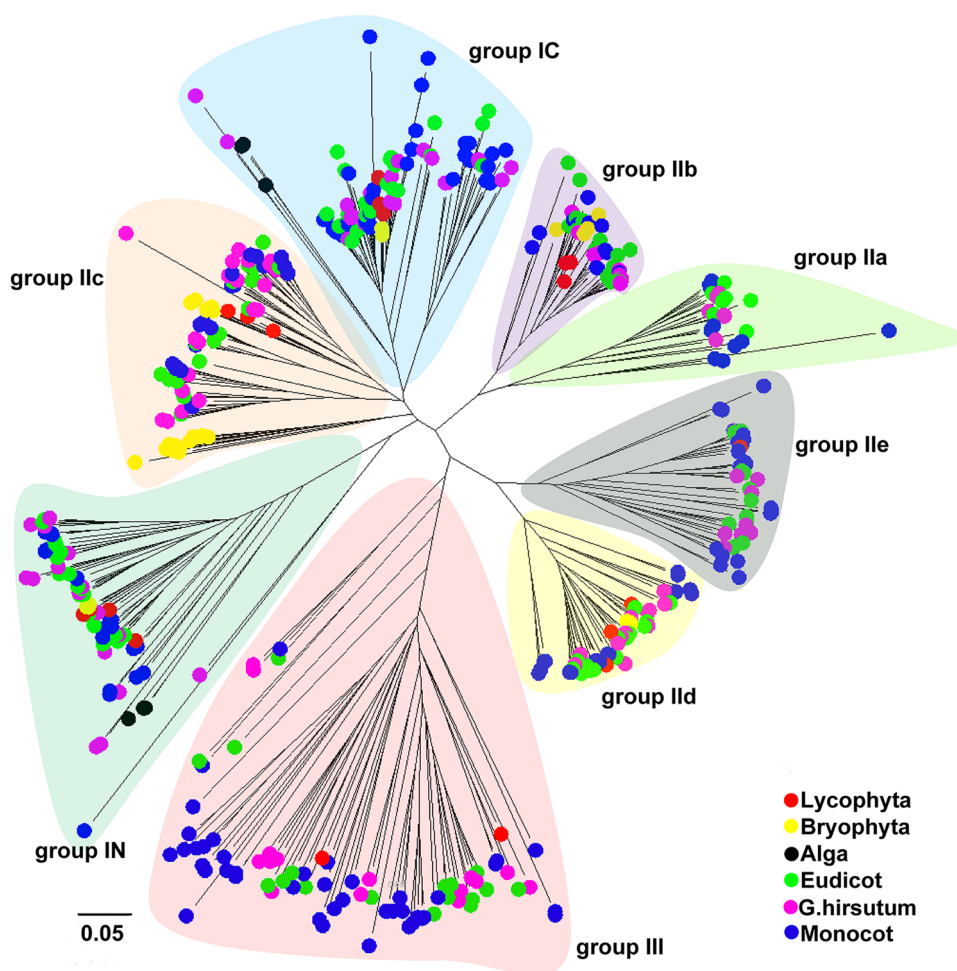


Fig. 3 Frequencies of simple sequence repeat (SSR) motifs in cotton WRKY genes. SSR motifs, with repeat units of more than six dinucleotides, four trinucleotides, or three tetranucleotides, pentanucleotides, or hexanucleotides were used as the search criteria. Most of the motifs comprised trinucleotides

GhWRKY gene expression patterns at different developmental stages and in specific organs

To understand the temporal and spatial expression patterns of these 102 *GhWRKY* genes in *G. hirsutum*, we used publicly available microarray data to assess the expression at

different developmental stages and under abiotic stress. We also used RNA-Seq data to analyze *GhWRKY* expression patterns during leaf senescence and anther development, and in specific tissues. The qRT-PCR method was used to confirm the expression data.

GhWRKY gene expression during leaf senescence and anther development, and in specific tissues

We used transcription profiling data to assess expression of *G. hirsutum* WRKY genes during the leaf senescence process in 15, 25, 35, 45, 55, and 65-day-old leaves. Analysis of these expression profiling data identified 55 *GhWRKY* genes expressed during leaf senescence (Fig. 4a).

Initially, most group IIC members had a very low expression level, and there was a significant increase in expression from 15 days onwards, especially for *GhWRKY39*, *104*, *101*, *93*, *96* and *108*. It is possible that group IIC genes might interact synergistically with other genes involved in the regulation of the aging process. The expression of only a few group I genes (for example, *GhWRKY18*) increased during leaf senescence. A group IIC member (*GhWRKY96*)

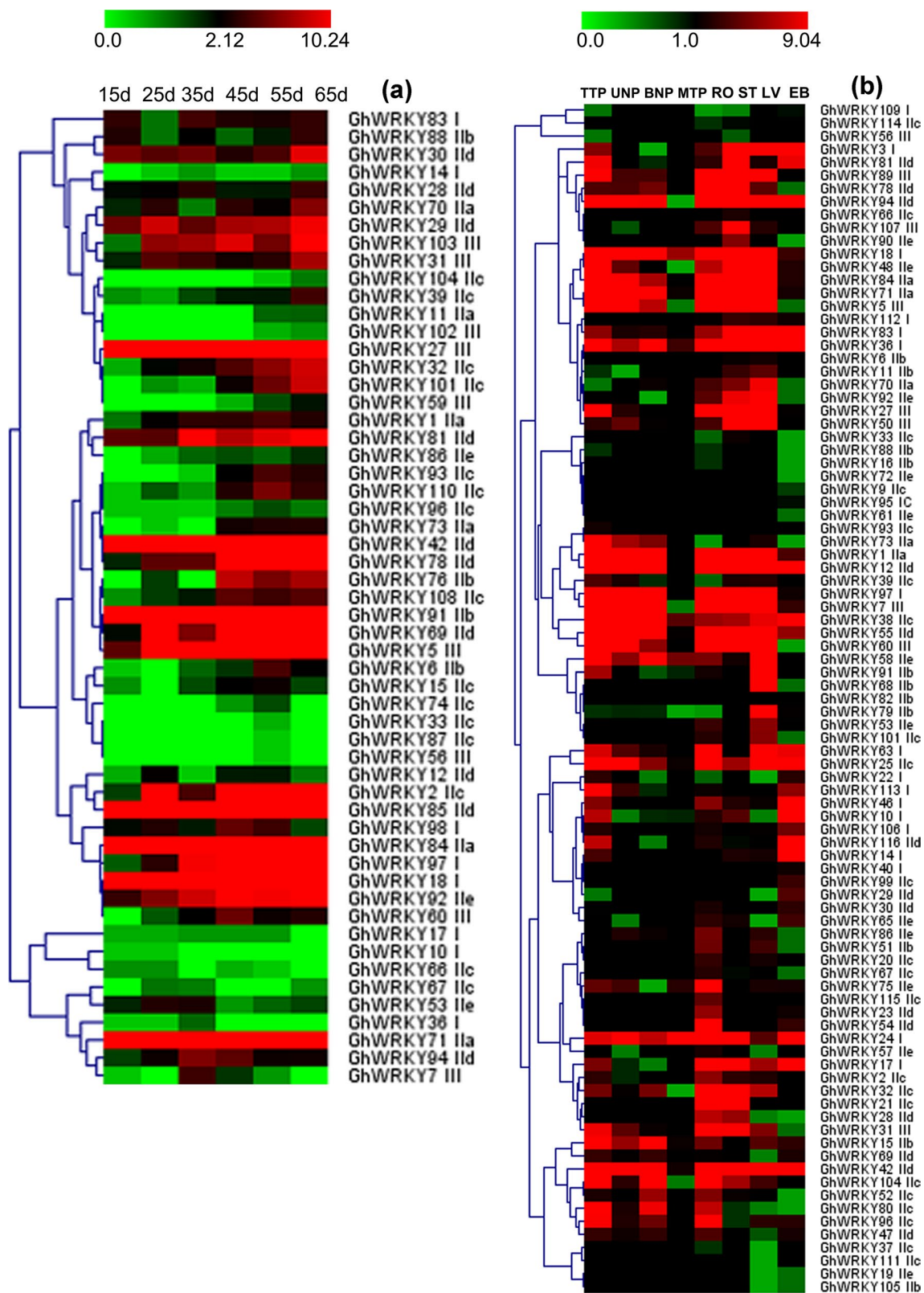


Fig. 4 *GhWRKY* gene expression during leaf senescence, anther development and in specific tissues. **a** The expression profiles of *GhWRKY* genes during leaf senescence from 15-, 25-, 35-, 45-, 55-, and 65-day-old leaves. The color bar represents the expression values, which have been normalized by the transcript per million clean

tags (TPM) algorithm. **b** The expression patterns of *GhWRKY* genes in TTP, UNP, BNP, MTP, root, stem, leaf and embryo. The color bar represents the expression values, which have been normalized by the reads per kilobase of exon model per million mapped reads (RPKM) algorithm

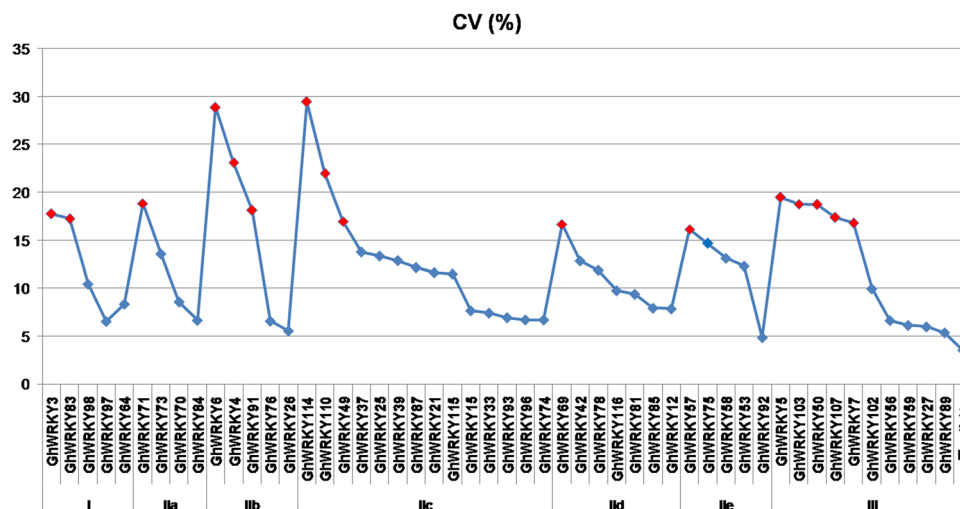


Fig. 5 Line graph showing the coefficient of variation (CV) of the signal intensities of 50 *GhWRKY* transcripts. A BLASTn search of the 102 *GhWRKY* genes in the CottonPlex database identified 50 *GhWRKY* genes expressed during fiber development at 0, 6, 9, 12, 19 and 25 dpa. Normalized signal intensities of the 50 *GhWRKY* tran-

scripts from 89 samples are displayed according to the CottonPlex database (GSE36228). Corresponding *GhWRKY* transcripts are presented on the x-axis, and ordered according to WRKY subgroups. A red diamond represents transcripts with CV values greater than 15 %

was upregulated in 15- to 45-day-old leaves (Fig. 4a) and was then downregulated. The genes of group IIc members (*GhWRKY32* and *101*), group IId member (*GhWRKY78*), group IIe member (*GhWRKY92*) and group III members (*GhWRKY5*, *59*) were upregulated throughout the senescence process. The expression levels of group III genes were relatively high and showed significant changes throughout the leaf senescence process (Supplementary data S1).

The expression levels of 91 *GhWRKY* genes (Fig. 4b) were detected in tissues from the developmental stages of anthers (TTP, UNP, BNP and MTP) (Ma et al. 2013), and in roots, stems, leaves and embryos in early inflorescence using the expression profiling data (Ma et al. 2012).

Some genes were specifically expressed in different tissues (Fig. 4b). Group IIb members (*GhWRKY68* and *79*) were leaf-specific, while group I members (*GhWRKY40* and *14*) were only expressed in embryos. Members from group IIc (*GhWRKY38* and *104*), group I (*GhWRKY36* and *24*), group IId (*GhWRKY94*) and group III (*GhWRKY5*) were constitutively expressed in all tissues.

Genes from group IId (*GhWRKY12*, *42*, *55* and *94*), group IIa (*GhWRKY1* and *71*), group I (*GhWRKY18*, *97* and *24*) and the group IIc (*GhWRKY38*) had very high expression levels during the anther development process from the TTP to BNP stage (Fig. 4b). Many *GhWRKY* genes showed high expression levels during anther development (Supplementary data S1); however, most of their functions have not yet been studied. Previously, we identified several WRKY genes that were differentially expressed during anther development, suggesting that WRKY genes

are components of a complex transcriptional network regulating anther development. Most of the *GhWRKY* genes had high levels of expression in the TTP and UNP stages (Fig. 4b). This result correlated with the genome-wide analysis of haploid male gametophyte development of *Arabidopsis*; the WRKY gene family was predominantly expressed at early stages of anther development (Honys and Twell 2004).

GhWRKY expression in fiber development

To investigate the expression patterns at different stages of *G. hirsutum* development, we calculated the coefficient of variation (CV) of each WRKY gene in 89 samples (Fig. 5). The CV values of these genes ranged from 4.86 to 29.42 %. Genes with CV values less than 15 % were considered to have low expression variability (Ishida et al. 2007). The reference gene encoding His3 (a histone protein) can be regarded as a constitutively expressed gene (Gou et al. 2007). By contrast, genes with CV values greater than 15 % are likely to be very important in cotton fiber development. There were 16 (31.37 %) *GhWRKY* genes expressed with a CV value greater than 15 %, and these genes are candidate genes for roles in cotton fiber development. The probe set information and expression data from the microarray are provided in Supplementary data S2.

To validate the importance of the genes with a CV value greater than 15 %, we performed quantitative real-time PCR (qRT-PCR) analysis during fiber development stages at 0, 5, 10, 15, 20 and 25 dpa from *G. hirsutum* species TM-1. High expression levels of these *GhWRKY*

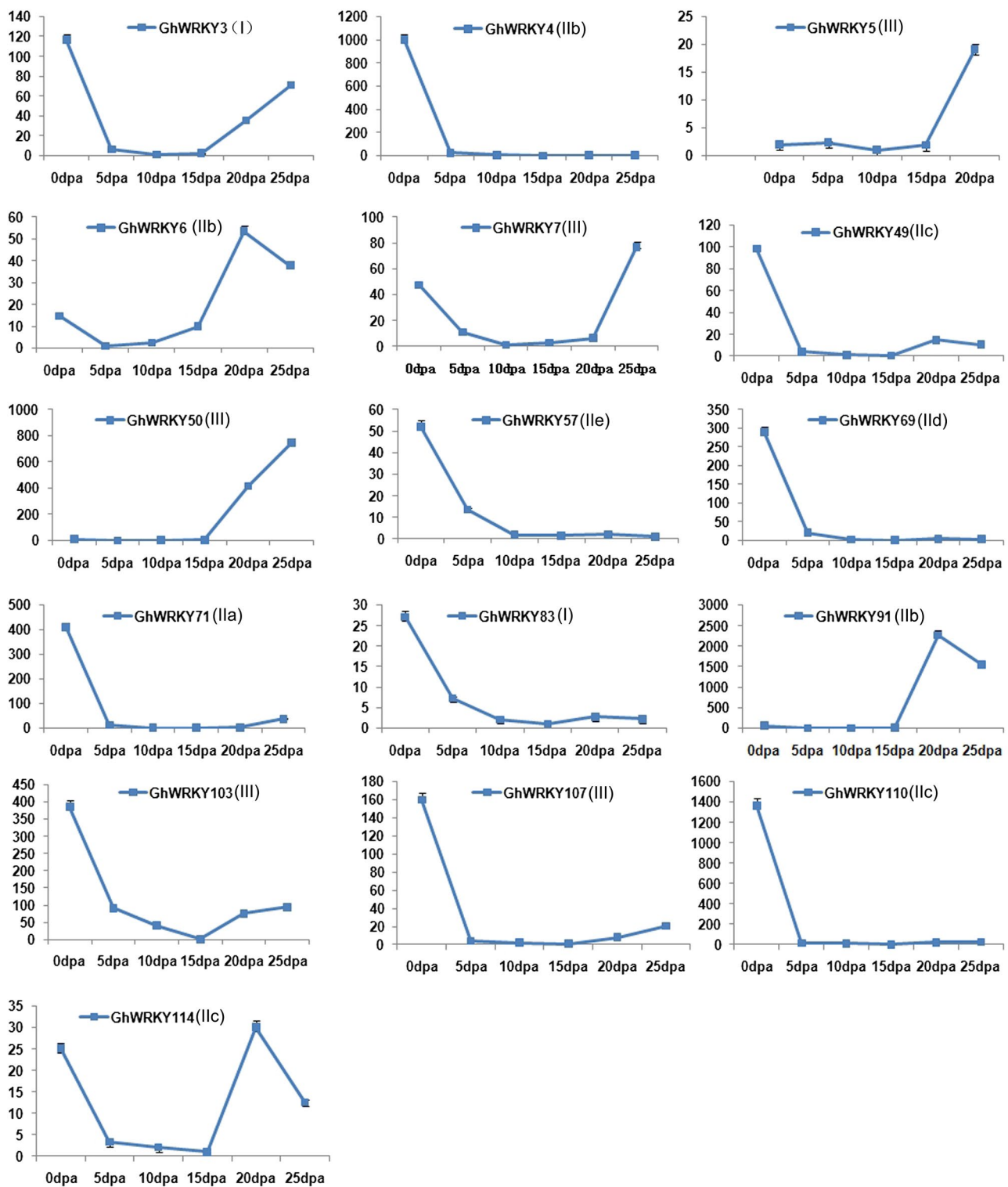
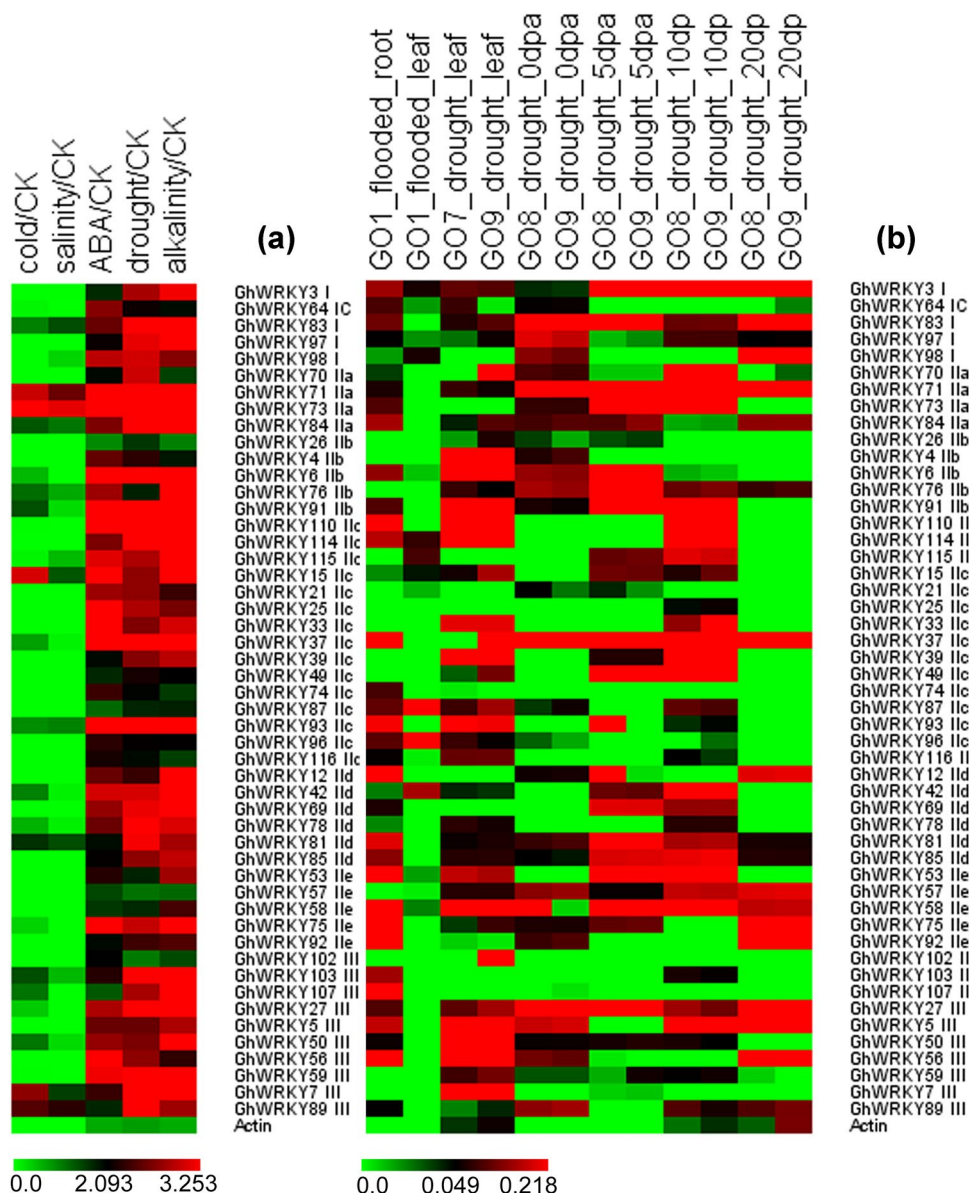


Fig. 6 Expression profiles of 16 *GhWRKY* genes assessed by qRT-PCR. The charts show the qRT-PCR-determined expression levels at 0, 5, 10, 15, 20 and 25 dpa

genes were seen at specific stages. *GhWRKY5*, 6, 50 and 91 were upregulated at the last stage of fiber development, which suggested that these genes might be involved in

fiber elongation and second cell wall formation (Hu et al. 2013). The initiation stage for fiber development occurs at 0 dpa, and this is also a crucial stage for fiber quantity

Fig. 7 Expression profiles of *GhWRKY* genes under abiotic stress. **a** Expression profiles of *GhWRKY* genes in *Gossypium hirsutum* roots and leaves under waterlog stress (GSE16467); and microarray expression data for leaf and fiber development stages (0, 5, 10 and 20 dpa) under drought stress (GSE18253, GSE29566, GSE29567 and GSE29810). **b** Expression patterns under cold, salinity, ABA, drought and alkalinity stresses (GSE50770). Both in **a** and **b**, the color bar represents the log₂ ratio compared with the control



development. *GhWRKY4*, 57, 71, 83, 107, 49, 69 and 110 showed very high expression levels at the 0 dpa stage, indicating that these *GhWRKY* genes have very important functions for initial fiber development (Fig. 6). The qRT-PCR primers are shown in Supplementary data S3.

Analysis of *GhWRKY* gene expression under abiotic stresses

We used microarray data to investigate the expression patterns of 50 *GhWRKY* genes in root and leaf tissues under waterlog stress (Fig. 7a; the expression data are shown in Supplementary data S4). Most *GhWRKY* genes had a higher expression level in the root than in the leaf under waterlog stress. Waterlogging results in lower levels of

oxygen in the plant root zone because of the low diffusion rate of molecular oxygen in water, which has an impact on energy metabolism. Response to low oxygen stress is also involved in energy metabolism in the roots and the highly expressed *GhWRKY* genes may play important roles in this complex biological process.

Under drought stress, members from group IIc (*GhWRKY33*, 39, 93, 110 and 114), group IIb (*GhWRKY4*, 6 and 91), group IIe (*GhWRKY58*) and group III (*GhWRKY5*, 7, 27, 50 and 56) had relatively high expression levels in the leaf. At 0 dpa, members from group III (*GhWRKY5* and 89), group IIb (*GhWRKY4*, 6 and 91), group IIa (*GhWRKY71*), group IIc (*GhWRKY37*) and group I (*GhWRKY83* and 97) had very high expression levels. Members from group IIb (*GhWRKY76*), group I (*GhWRKY3* and 83), group IIa

(*GhWRKY73*), group IIc (*GhWRKY49* and 37) and group III (*GhWRKY27*) were specifically expressed at 5 dpa. At 10 dpa, members from group IIc (*GhWRKY49* and 110), group IIb (*GhWRKY91*), group IId (*GhWRKY42*) and group IIe (*GhWRKY53* and 58) had enhanced expression. At 20 dpa, members from group IIa (*GhWRKY71*), group IIc (*GhWRKY37*), group IIe (*GhWRKY92*) and group III (*GhWRKY5*) were highly expressed. Under drought stress, these highly expressed genes seem to be more sensitive during fiber development. Therefore, specifically upregulated *GhWRKY* genes may directly or indirectly control certain stages of fiber development under drought stress.

Under cold, salinity, ABA, drought and alkalinity stress (Fig. 7b; the expression data are shown in Supplementary data S4), members from group IId (*GhWRKY81*), group IIc (*GhWRKY15*), group III (*GhWRKY7*, 89) and group IIa (*GhWRKY71*, 84 and 73), responded to all five kinds of abiotic stress, with expression level changes of more than two-fold compared with the control. This further suggested that these common upregulated *GhWRKY* genes possibly participate in cross-talk between signaling pathways to regulate these five kinds of stresses.

We also noticed that most *GhWRKY* genes were induced by each of ABA, drought and pH stress, while cold and salinity (especially) caused expression changes in fewer genes. This suggested that there were more *GhWRKY* genes involved in ABA, drought and pH stresses compared with cold and salinity in cotton seedlings. It strongly indicated that *GhWRKY* genes have very important functions in cotton's adaptation to ABA, drought and pH stresses.

Discussion

WRKY genes in cotton

Cotton plays an important role in the global economy. Improvements in quality characteristics and production can be achieved by molecular breeding and conventional cross breeding. Genetic and molecular analysis of functional genes will be very important for cotton molecular breeding. WRKY proteins are one of the most important families of transcription factors in cotton (Brand et al. 2013). Cai et al. (2014) recently reported a genome-wide analysis of the WRKY gene family members in *G. raimondii* based on sequence information from Paterson et al. (2012). They discussed the number, distribution on chromosomes, and structure of genes and presented a basic analysis of the conserved motifs in WRKY proteins. In this study, 116 *GrWRKY* genes were identified directly from the sequenced genome of *G. raimondii* and 102 *GhWRKY* genes were identified in *G. hirsutum* by homology-based cloning methods. We then used large-scale methods to measure the

temporal and spatial expression patterns of these genes. To understand the evolutionary relationships of WRKY family genes in plants, WRKY proteins from algae, glaucophyta, chlorophyta, bryophytes, lycophyta, monocots and eudicots were included in phylogenetic analyses. These studies elucidated the classification and phylogenetic relationships among different subgroups of cotton WRKY genes. To understand the relationships among highly similar orthologous *GrWRKY* and *GhWRKY* sequences, we analyzed SSRs between *G. raimondii* and *G. hirsutum*. We also determined their locations on chromosomes, which provide evidence to explain the existence of large numbers of WRKY gene family members (see below). As an important family of regulatory genes it is important to understand the evolution and the molecular and biological functions of WRKY genes in these two cotton species.

Gene families arise through segmental duplications of chromosome regions, resulting in a scattered pattern of occurrence, or through tandem amplification, resulting in a clustered pattern (Schauser et al. 2005). To analyze gene clusters and duplication events of WRKY genes in *G. raimondii*, we defined a cluster as the occurrence of two or more *GrWRKY* genes located within 40 open reading frames (Meyers et al. 2003). Tandem duplication occurs when two closely related *GrWRKY* genes are located within the same chromosome region and are fewer than 20 genes apart (Xu et al. 2009). We identified nine gene clusters, which could be classified into three types. The first comprised four gene clusters that form a monophyletic tandem duplication together with intervening non-WRKY genes. These clusters included members of subgroups IIa, IIc, IId and III. The second comprised three gene clusters composed of a mixture of subgroups IIc and IIe. The third type comprised a mixture of the two previously mentioned groups (Supplementary Table S5). The number of distinct gene clusters in *G. raimondii* (9) is less than that in maize (20). The WRKY gene family in *G. raimondii* seems to have mainly evolved from segmental duplication followed by tandem amplifications. In *Arabidopsis*, it was confirmed that the *AtWRKY* gene family resulted from a few tandem duplications and a moderate number of segmental duplications (Cannon et al. 2004). *G. raimondii* has the least-repetitive DNA sequence of the *Gossypium* members (Wang et al. 2012), which might explain why there is a relatively low number of tandem repeats in the *GrWRKY* gene family.

The high similarity between *GrWRKY* and homologous *GhWRKY* gene sequences makes it difficult to distinguish them; therefore, SSR analysis proved very useful to analyze the evolutionary differences and classification of the WRKY sequences of these two cotton species. The classification of SSR motifs into three groups provides a foundation for further analysis of evolutionary sequences between these two species (Kantety et al. 2002).

Phylogenetic analysis of the WRKY proteins from algae (*C. merolae*, *Ch.sp.NC64A*, *Ch. reinhardtii*, *V. carteri*), bryophytes (*P. patens*), lycophyta (*S. moellendorffii*), monocots (*Z. mays* and *O. sativa*), and eudicots (*P. trichocarpa*, *A. thaliana* and *G. hirsutum*), suggested that WRKY proteins can be classified into group I, group IIa + IIb, group IIc, group IId + IIe and group III. This classification of WRKY groups is the same as that suggested by Zhang and Wang (2005); however, it differs slightly from the traditional classification of Eulgem et al. (2000). In addition, the phylogenetic tree implied that group IIa and IIb are closely related, with no alga, bryophyte or lycophyta domain classified into group IIa. Thus, we inferred that group IIa may have evolved from group IIb. The same was true for group IIe and IId. The WRKY transcription factor family investigated in this study showed an increasing number of WRKY family members from algae to angiosperms. This expansion co-evolved with the increased complexity of the plant body. As an important transcription factor, rapidly increasing its family members may have optimized plant adaptability and enabled the establishment of signal transduction webs in times of adversity (Kumar et al. 2011).

Expression patterns of *GhWRKY* genes during cotton development

Four *GhWRKY* genes (*GhWRKY34*, 45, 77 and 100) showed no detectable expression, as tested by screening of the microarray expression data and from the expression profiling data. They may be expressed at a very low level in the conditions and tissues tested, the probe set information might be limited, or they are expressed only in specific conditions (Ling et al. 2011).

Phylogenetic analysis suggested that group I genes are the ancestors of the other WRKY groups present in ancient organisms and are prevalent in plants (Zhang and Wang 2005). All plants analyzed, from algae to *G. hirsutum*, have WRKY proteins that could be classified into group I (Fig. 2). These genes were expressed in almost every tissue (Figs. 4, 5 and 6); therefore, they are more likely to be constitutively expressed (Zhang and Wang 2005). For example, group I members (*GhWRKY3*, 83 and 97) were expressed not only under abiotic stress and in fiber development stages, but also were highly expressed during leaf senescence, anther development, and in roots, stems, leaves and embryos. *GhWRKY3* (CV = 17.71) was stably expressed during the senescence process and *GhWRKY83* (CV = 17.22) was highly upregulated in 15- to 45-day-old leaves, after which they were downregulated. *GhWRKY18* also showed higher expression during anther development from the TTP to MTP stages. This may be because anther development stages from TTP to MTP involve not only active differentiation activity, but also programmed

cell death. Highly expressed genes may play a regulatory role in leaf development (Li et al. 2012). For example, *GhWRKY18* is homologous with *AtWRKY33*, a senescence-associated gene (Lippok et al. 2007), and group I members also have a very important role in regulating leaf senescence.

Group IIa members of *Arabidopsis* (*AtWRKY18*, *AtWRKY40*, and *AtWRKY60*) have partially redundant roles in response to the hemibiotrophic bacterial pathogen *Pseudomonas syringae* and the necrotrophic fungal pathogen *Botrytis cinerea*; *AtWRKY18* has a more important role than the other two (Shen et al. 2007). Therefore, *GhWRKY* genes from group IIa may also have redundant roles in response to these abiotic stresses.

Group IIc members of *Arabidopsis* (*AtWRKY8*, 48, 50, 51 and 57) respond to bacteria, fungi, and jasmonic acid (JA)- and SA-mediated signaling pathways (Tang et al. 2013). *AtWRKY23* (group IIc) is an auxin-inducible gene that can be induced by *Heterodera schachtii*. It acts downstream of the Aux/IAA protein SLR/IAA14 (Grunewald et al. 2008). *BnWRKY28* (IIc) and *BnWRKY45* (IIc) can be induced by infection with *Sclerotinia sclerotiorum* or by ethylene (Yang et al. 2009). In this study, most group IIc members had very low expression levels, but showed significant changes in expression during leaf senescence. It is possible that group IIc genes might interact synergistically with other genes involved in the regulation of the aging process. During fiber development, three group IIc genes *GhWRKY110* (CV = 21.93), *GhWRKY49* (CV = 16.94) and *GhWRKY114* (CV = 29.422) showed large-fold expression changes compared with *GhACT3*, another reference gene in *G. hirsutum*. It is possible that many group IIc members not only respond to biotic stress, but also play an important role in fiber development and leaf senescence.

Many *Arabidopsis* group III members, including *AtWRKY30*, 53, 54, and 70 have been identified as senescence regulators (Besseau et al. 2012). Comparative analysis of orthologs helped to determine the evolutionary relationships among gene family members and predicted potential functions of putative proteins. The most homologous gene of *AtWRKY53* (III) is *GhWRKY27* and the most homologous gene of *AtWRKY54* and *AtWRKY70* (III) is *GhWRKY102*. *AtWRKY53* is expressed very early in the leaf senescence process and works as a positive regulator of senescence (Woo et al. 2010). Coincidentally, *GhWRKY27* (III) was highly upregulated during the senescence process, suggesting a similar function for these two homologous genes. Therefore, these *GhWRKY* genes might play very important roles in cotton leaf senescence.

Under drought stress, group III members (*GhWRKY5*, 7, 50 and 56) showed relatively high expression levels in the leaf. These group III members were expressed throughout all fiber developmental stages under drought stress,

suggesting that group III members have very important roles in response to abiotic stress. Only two lycophyta WRKY proteins were classified into group III and no alga, bryophyte or lycophyta WRKY proteins were classified into groups IIa or IIe (Fig. 2); most *GhWRKY* genes from these groups have very high expression levels under stressful conditions (Fig. 7). During the evolutionary process, group III *GhWRKY* genes may have developed very important roles in response to stress stimuli (Fig. 7).

Under normal conditions, most *GhWRKY* genes are expressed at low levels in all developmental processes, while only a few genes are highly expressed in specific organs or developmental processes. This suggests that some *GhWRKY* genes show stage- or tissue-specific expression, which is similar to findings for the *ZmWRKY* family members (Wei et al. 2012). Under abiotic stress, most *GhWRKY* genes show very high expression, implying that WRKY genes function in the stress response. In this study, we analyzed the expressions of *GhWRKY* genes during fiber development (0, 6, 9, 12, 19 and 25 dpa), leaf senescence, anther development, and in tissues (roots, stems, leaves and embryos). Expression patterns in roots and leaves under waterlog stress, drought stress and during leaf and fiber development (0, 5, 10 and 20 dpa) were also investigated.

The results showed that most of the *GhWRKY* genes have high expression divergence in fiber development, leaf senescence and in response to abiotic stress, including waterlog stress and drought stress. In this regard, *GhWRKY* genes might have diverse functions in cotton development. SSRs may substantially increase the rate of duplication of DNA segments (Alkan et al. 2011); the high frequency of SSRs in *G. raimondii* and *G. hirsutum* might imply increased evolution of *GhWRKY* gene family members. According to our analysis of the chromosomal distribution of *GrWRKY* genes, it is possible that *GrWRKY* genes mainly evolved from segmental duplication followed by tandem amplifications. Segmental duplication and tandem duplication have contributed significantly to the expansion of plant gene families (Zhang and Gaut 2003). It is thought that during vascular plant evolution, tandem duplications tend to be involved in stress responses (Rizzon et al. 2006). The many WRKY members in cotton, induced by adaptive duplication, might have led to an increased sensitivity to stress. It will be very interesting to analyze the transcription regulated by key WRKY proteins in *G. hirsutum* because of cotton's global economic importance. These *GhWRKY* genes with positive effects on fiber development and responses to abiotic stress promote fiber yield and quality, which are important cotton economic traits. In conclusion, this study has not only provided an updated analysis of evolutionary relationships and chromosomal distributions of WRKY family genes in cotton species. It has also analyzed the expression levels of these genes under diverse

stresses and developmental processes. These data represent an important reference and foundation for further studies of the functions of WRKY proteins in cotton species.

Acknowledgments We thanks for the National Basic Research Program of China (Grant No. 2010CB126006) and the China Agriculture Research System (Grant No. CARS-18) providing the financial support for this project. We are grateful to the researchers who submitted the microarray data to the public expression databases. We are also grateful to all of the members of our laboratories who completed the expression profiling. We also thanks for EVans Ondati to help us revise the language.

Conflict of interest The authors declare that they have no conflict of interest.

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